

Taskora — SaaS Project Study & Architecture Manual

Tagline: Work Together. Finish Faster. | Version: 2.0.0 Production Ready

1. Executive Summary & Core Vision

Taskora is an enterprise-grade full-stack SaaS project management platform engineered to streamline sprint execution, workspace delivery, and developer productivity. Built with a modern glassmorphic UI, custom branding, and a Pomodoro **Focus Mode** telemetry engine, Taskora offers software teams a cohesive environment to collaborate, track deliverables, and finish projects faster.

2. Complete Technology Stack

Layer	Technology / Framework	Purpose & Key Libraries
Frontend Core	React 18 + Vite	Component-based UI engine with 500ms HMR dev server and optimized Rollup production builds.
Styling & Brand	Tailwind CSS v3	Custom design system tokens (#FF6B00 primary accent, #111827 dark slate, rounded-2xl 16px corners, glassmorphism overlays).
SPA Routing	React Router DOM v6	Declarative single-page routing (/overview, /workspace, /sprint, /members, /insights, /focus, /profile, /login, /register).
HTTP Client	Axios	Axios instance configured with JWT Bearer request interceptors and centralized response error handling.
Data Visualization	Recharts v2	Interactive SVG area charts, bar graphs, and pie charts featuring custom high-contrast Tooltip rendering.
Backend Server	Node.js + Express.js	Modular REST API architecture with controllers, middleware, CORS, and JSON body parsing.
Auth & Security	JWT + bcryptjs	JSON Web Tokens stored in headers/localStorage, password hashing, and Role-Based Access Control (RBAC).
Database Layer	PostgreSQL / SQLite	Dual-mode database adapter: production PostgreSQL pool with embedded SQLite zero-config dev fallback.

3. Folder Structure & Modular Architecture

The project strictly adheres to feature-based modular organization on the client and MVC on the server:

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Frontend Structure (client/src/):
• features/: Domain modules (auth/, overview/, workspace/, sprint/, members/, insights/, focus/, profile/)
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- **shared/**: Reusable UI components (Modal.jsx, Badge.jsx, CustomTooltip.jsx, EmptyState.jsx)
- **layouts/**: AppLayout.jsx, Sidebar.jsx (React Router NavLink), TopNav.jsx (Global search & notification dropdown)
- **services/**: api.js (Axios API client with request/response interceptors)
- **context/**: AuthContext.jsx, ThemeContext.jsx (Light/Dark mode), NotificationContext.jsx (Toast alerts)

Backend Structure (server/src/):

- **controllers/**: authController, workspaceController, taskController, memberController, insightsController, userController
- **routes/**: Express route definitions (/api/auth, /api/workspaces, /api/tasks, /api/members, /api/insights, /api/user)
- **middleware/**: authMiddleware.js (JWT verify Token & authorizeRoles middleware)
- **config/**: db.js (Dual DB Query wrapper), schema.js (5 table DDL definitions)
- **seed/**: seed.js (Populates realistic users, workspaces, sprint items, members, activity logs)

4. Implemented Modules & Application Features

- **Overview Dashboard**: Presents 4 key KPI cards (Active Workspaces, Tasks Completed Today, Team Efficiency, Productivity Score Gauge 95%), Weekly Progress Area Chart, Deadline Tracker, and Recent Activity Stream.
- **Workspace Module**: Manages the product pipeline across 4 stages: **Idea** → **Building** → **Testing** → **Delivered**. Features search, priority filtering, grid vs. table view toggle, and Create/Edit/Delete modals.
- **Sprint Board**: Interactive HTML5 Drag-and-Drop Kanban workflow covering **Backlog**, **In Progress**, **Review**, **Completed** columns. Instant optimistic state sync, task priority badges, and task details drawer modal.
- **Members Directory**: Team roster supporting role assignments (**Admin**, **Project Lead**, **Developer**, **Designer**), activity status indicators (*Active*, *Away*, *In Focus*), and member invitation/removal.
- **Insights Telemetry**: Analytics engine rendering Recharts Task Completion Rate pie chart, Workspace Progress bar chart, and Daily Productivity trends with high-contrast CustomTooltip styling.
- **Focus Mode (Unique Feature)**: Built-in **25-minute Pomodoro Focus Timer** bound to active sprint tasks. Includes play/pause/reset controls, progress ring, and session productivity metrics logger.
- **Profile & Settings**: User profile editor, password modification form, email notification toggles, and seamless dark mode switch.

5. Database Schema & REST API Specifications

Table Name	Key Columns	Relationship / Purpose
users	id, name, email, password, role, avatar, title, bio, theme, created_at	Stores authentication credentials (hashed password via bcryptjs) and user preferences.
workspaces	id, name, description, priority, status, start_date, end_date, created_by	Stores project workspace pipelines (Idea, Building, Testing, Delivered).
work_items	id, title, description, priority, due_date, status, workspace_id, assigned_to	Sprint Kanban tasks linked to workspaces and assigned users (Backlog, In Progress, Review, Completed).
team_members	id, user_id, workspace_id, role, activity_status, joined_at	Links users to workspaces with specific roles and real-time activity status (Active, In Focus).
activity_logs	id, user_id, user_name, user_avatar, action, target, timestamp	Audit log trail recording task moves, workspace creation, and member invitations.

REST API Endpoints Reference:

Method	Endpoint	Auth	Description & Payload
POST	/api/auth/register	Public	Registers new user, hashes password, returns JWT token.
POST	/api/auth/login	Public	Authenticates email/password, returns signed JWT token.
POST	/api/auth/demo-login	Public	Single-click demo login for Admin, Project Lead, or Developer roles.
GET	/api/workspaces	JWT	Lists workspaces with status/priority filtering and computed progress %.
POST	/api/workspaces	JWT	Creates new workspace with Idea, Building, Testing, or Delivered stage.
GET	/api/tasks	JWT	Fetches sprint work items with workspace and assignee details.
PUT	/api/tasks/:id	JWT	Updates task stage during drag-and-drop or details edit.
GET	/api/members	JWT	Lists team members with active task counts and efficiency scores.
GET	/api/insights	JWT	Aggregates telemetry data for Recharts graphs and KPI metrics.

6. Key Full-Stack Developer Takeaways & Best Practices

- **Axios Request Interceptor:** Automatically injects the JWT token stored in localStorage into the Authorization: Bearer <token> HTTP header for every API request, avoiding repetitive code.
- **Recharts High-Contrast Tooltips:** Default Recharts tooltips can render dark text on dark backgrounds. By building a custom React CustomTooltip component, we enforce 100% white text contrast (#FFFFFF) on elevated slate boxes in both Light and Dark modes.

- **HTML5 Drag and Drop with Optimistic UI:** When moving a task card across Kanban columns, the UI state updates instantly on drag release for 60fps smoothness, while an async PUT `/api/tasks/:id` request updates the database in the background.
- **Pomodoro Focus Engine:** Stateful 25-minute timer using React `useEffect` interval loops, bound to active task state, tracking total focus minutes and completed sessions.
- **Dual Database Adapter Pattern:** The backend database wrapper (`server/src/config/db.js`) seamlessly connects to PostgreSQL in production environments while automatically providing an embedded SQLite file (`taskora.db`) for instant local zero-setup dev execution.